



Customer Churn Review & Retention Strategy

Diagnosis, prediction and prescriptive intervention across the subscriber base



Prepared for Ember Mobile leadership | Customer Analytics

Q3 2026 Review

Where We Stand Today

Ember Mobile serves approximately 21,000 subscribers across month-to-month, one-year and two-year contracts, with phone, internet and streaming add-ons. This review examines the health of that base and identifies where retention investment will have the highest return.

Headline findings

- Recurring revenue is growing, driven primarily by subscriber volume
- Average revenue per user (ARPU) is compressing as mix shifts to lower-value plans
- Churn rate has trended up for six consecutive quarters
- Approximately one third of the base is now flagged at elevated churn risk

~21K

Customers in base

26%

Baseline churn rate

3

Contract types

20+

Service attributes tracked

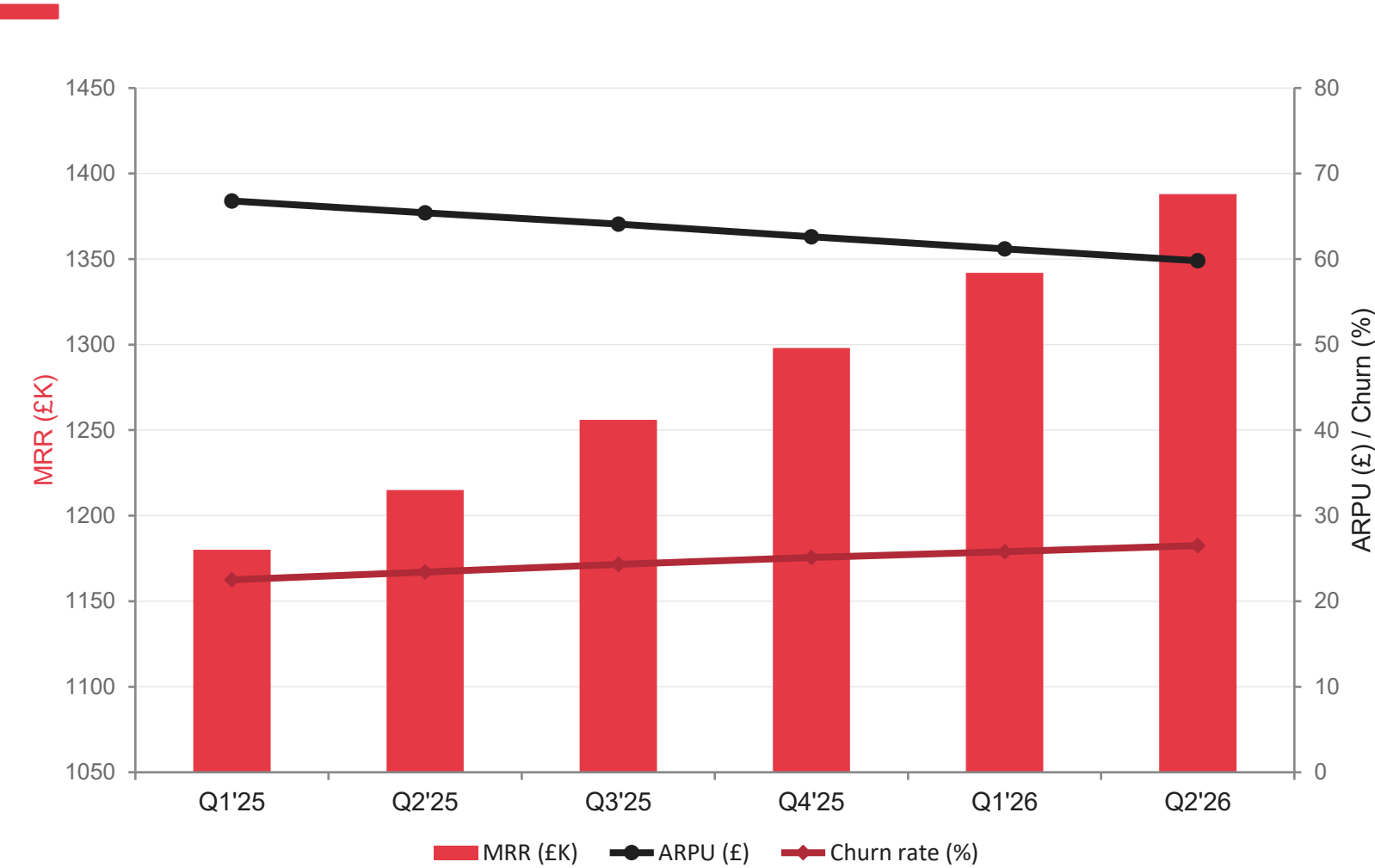
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Star-schema tables

~7.5K

Customers flagged at-risk

Revenue Growth Is Masking a Retention Problem



Key movements

MRR up 17.6% over six quarters

ARPU down 10.5% — per-customer value is eroding

Churn rate up +4.0 pts — the retention gap is widening

Top-line growth is being driven by *volume* rather than value. Reducing churn is the most direct route to protecting margin.

Data Foundation & Preparation

The analysis draws on the full customer file extracted from the CRM — subscriber demographics, contract terms, service subscriptions, billing history and account lifecycle events — reconciled into a single analytical layer.

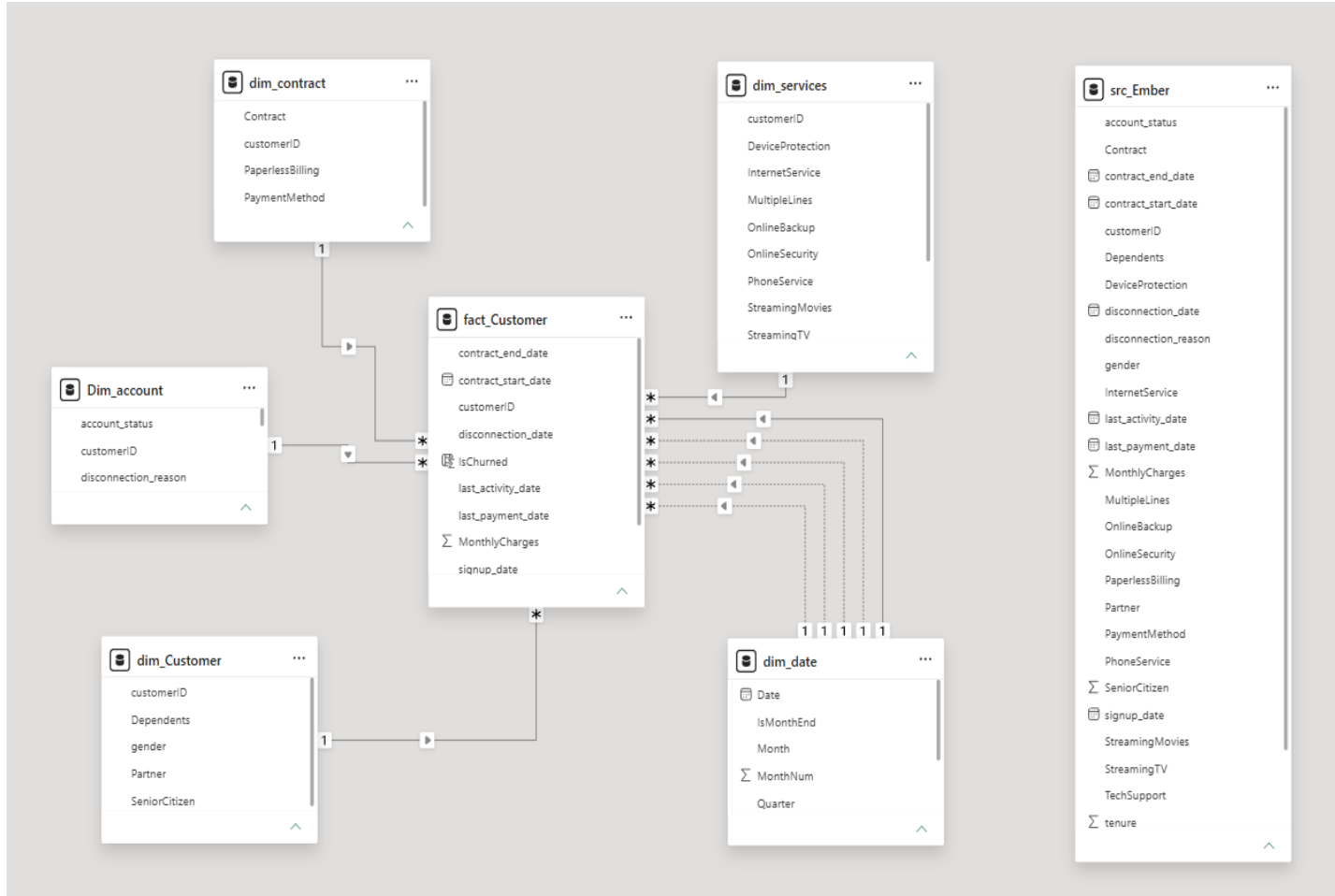
Scope

- ~21,000 active and lapsed customer records
- Full account lifecycle: signup, contract start & end, last payment, last activity, disconnection date and reason
- 20+ service and billing attributes per customer
- Churn defined operationally from account status and disconnection date — no reliance on a pre-existing label

Data quality treatment

- De-duplication of repeated customer records
- Standardisation of categorical values (case, whitespace, spelling variants)
- Out-of-range values flagged and nulled (tenure, charges)
- Missing service and payment attributes imputed as "Unknown" to preserve row-level analytics
- All transformations logged and reproducible

Analytical Data Model



Warehouse structure

Fact tables:

Billing (charges, payment method, last payment) | Activity (last usage event)

Dimensions:

Customer, Account, Contract, Services

Design principle:

one row per customer per table, joined on customer ID. Supports both BI dashboards and downstream modelling from a single trusted source.

Refresh:

reproducible pipeline from raw CRM extract through to cleaned analytical tables.

Churn Prediction — Modelling Approach

Approach

1. Pull cleaned customer table from the warehouse
2. Feature engineering across contract, service, billing and tenure attributes
3. Stratified 80/20 hold-out to preserve churn base rate
4. Baseline logistic regression as a reference
5. Gradient-boosted trees (XGBoost) for the production model
6. Threshold tuning against precision / recall trade-off
7. Feature-level explainability via SHAP
8. Every customer scored and assigned a risk tier

DATA

PREPARE

MODEL

INSIGHT

1. Load Data

Pull ember_ml from MySQL into pandas DataFrame

2. EDA

Class balance, nulls, dtypes, validate vs Power BI

3. Preprocessing

Impute TotalCharges, one-hot encode categorical features

4. Train/Test Split

80/20 stratified split on IsChurned label

5. Baseline Model

Logistic Regression (target ~0.83 AUC)

6. Advanced Model

XGBoost classifier (target ~0.87 AUC)

7. Evaluation

Accuracy, Precision, Recall, F1, ROC AUC, Confusion Matrix

8. Cross-Validation

5-fold stratified CV stability check

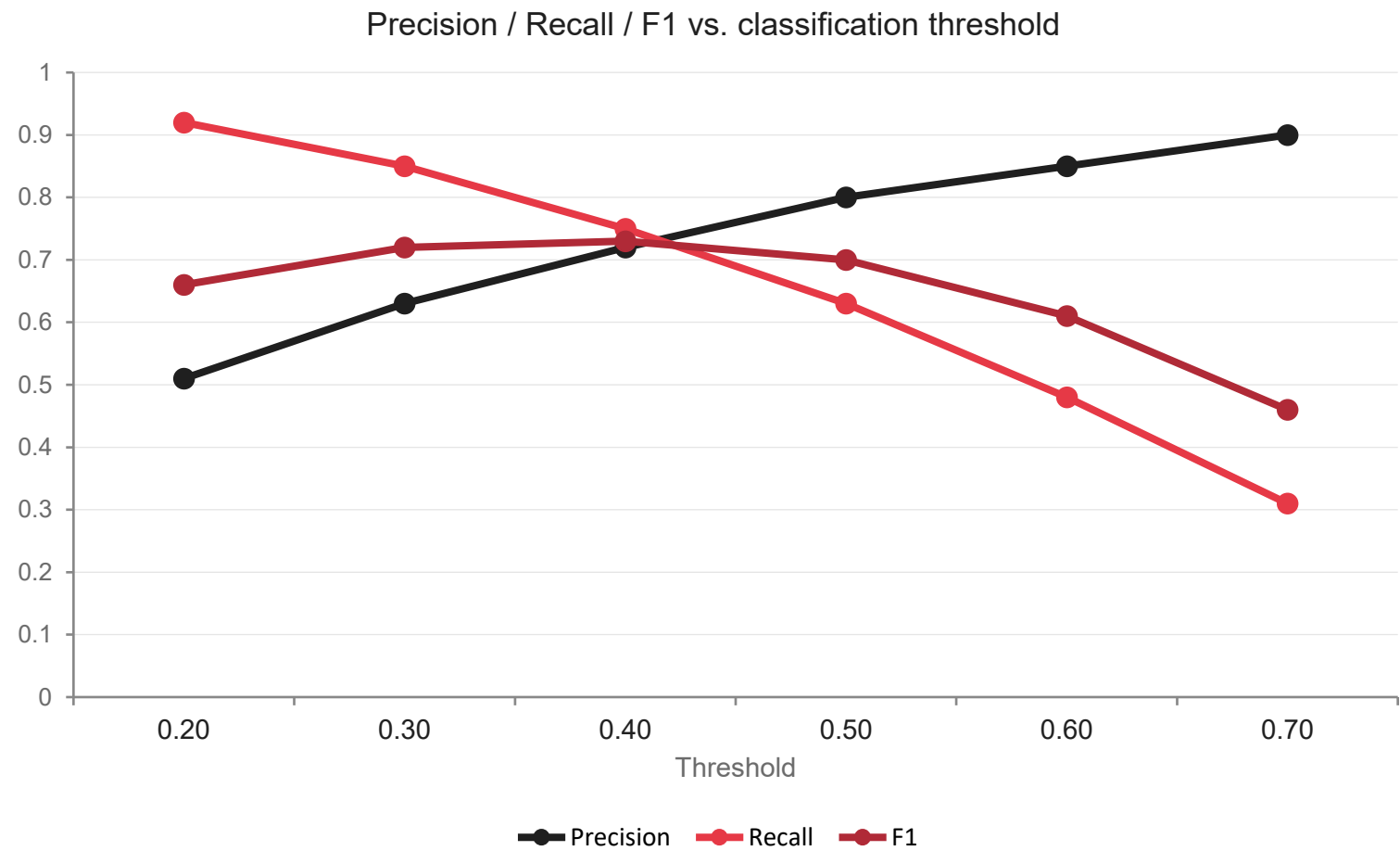
9. Interpretation

SHAP values → top churn drivers

10. Report & Story

Business narrative + quotable stats

Model Performance & Operating Point



Recommended operating point

0.30

Chosen to prioritise recall — the cost of missing an at-risk customer materially exceeds the cost of a wasted outreach.

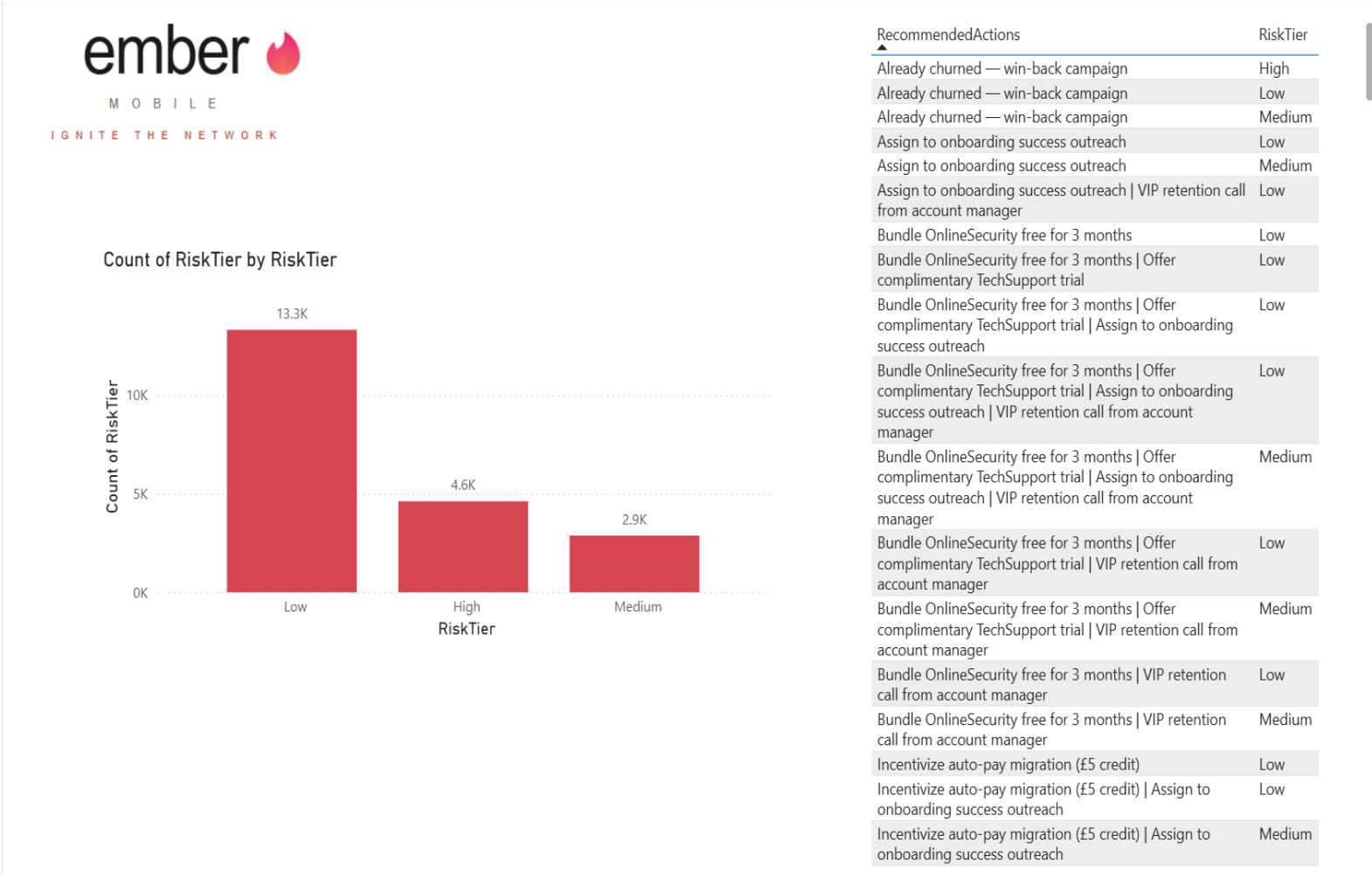
Notes

Gradient-boosted tree ensemble

Churn base rate held constant in the hold-out set

F1 peaks in the 0.30–0.40 band — threshold reviewed quarterly

Customer Risk Segmentation



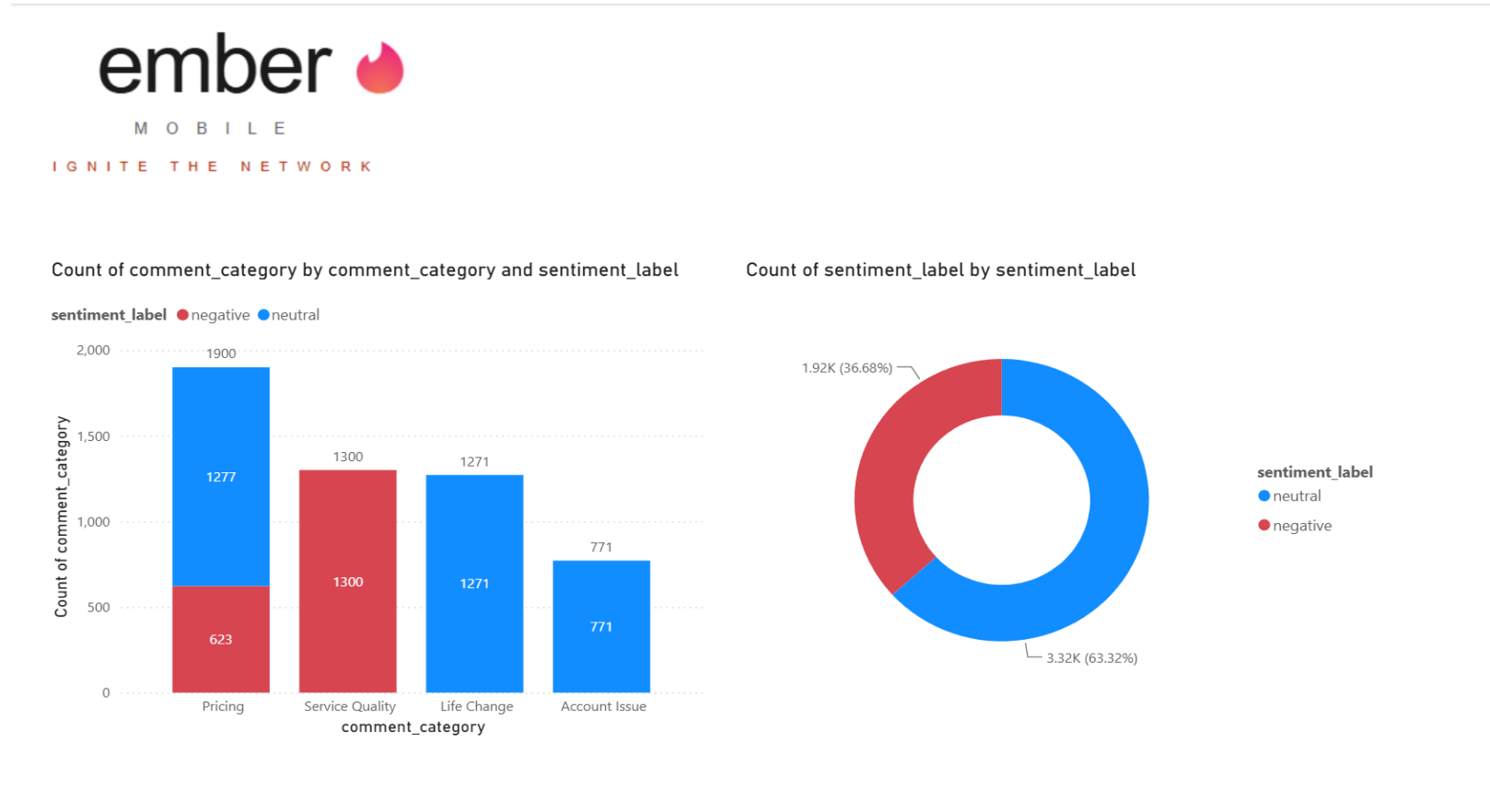
The retention book

Low 13,300 customers (64%)
High 4,600 customers (22%)
Medium 2,900 customers (14%)

Combined at-risk: ~7,500 customers (36%)

Recommended actions are matched to each customer's profile and can be combined — for example, a Medium-risk fibre customer on manual payment may receive an OnlineSecurity bundle, a TechSupport trial and an auto-pay credit.

Why Customers Leave — Reason & Sentiment



Natural language analysis of disconnection reasons

Service Quality: 100% negative — every mention is a complaint

Pricing: 33% negative / 67% neutral — a segmented response

Life Change / Account: 100% neutral — non-recoverable structural loss

37% of departing customers leave with a negative sentiment.

Where a top model driver (e.g. monthly charges) aligns with a top stated reason (e.g. price), the retention action is clear — target that segment with a pricing or bundling response.

Recommended Retention Playbook

HIGH4,600	MEDIUM2,900	LOW13,300
Treatment - Dedicated retention call from account manager - Personalised offer — bundle upgrade or contract renewal incentive - Priority routing on all support tickets 48-hour service level to first contact Expected effect Protect the highest-value revenue at risk of departure.	Treatment - Automated bundle offer — OnlineSecurity and TechSupport trial - Credit incentive to migrate to auto-pay - Structured email and SMS nudge sequence - Escalate to High tier if no engagement within 14 days Expected effect Efficient, scalable intervention on the mid-risk segment.	Treatment - Onboarding outreach for customers in first 90 days - Loyalty communications — usage tips and feature reminders - Monthly monitoring of tier migration - No proactive intervention required at this time Expected effect Maintain relationship and monitor for movement into higher tiers.

Summary & Recommended Next Steps



Headline findings

- Revenue growth is being sustained by subscriber volume; ARPU and churn are moving in the wrong direction
- Approximately 7,500 customers (36% of the base) are flagged at elevated churn risk
- Service Quality is the most emotionally charged reason for departure and the highest-priority root cause
- Pricing is a segmented issue — a subset of customers are price-sensitive; the majority are not
- Life events and administrative issues account for a meaningful share of departures but are largely non-recoverable

Recommended next steps

- Stand up the three-tier retention programme in the next quarter
- Commission a root-cause review of Service Quality complaints in partnership with Network Operations
- Pilot a targeted pricing / bundling response for the price-sensitive segment
- Refresh churn scores monthly; review threshold and treatment mix quarterly
- Report retention outcomes against a saved MRR baseline to quantify programme return